3 User interface

This chapter covers the basic know-how for building an Android user interface. At the end of this chapter, you'll be able to create an activity and an XML layout for the activity, and you'll learn how to connect it to Java. We'll also cover Views (or Widgets) and Layouts, and learn how to handle Java events such as clicks. You'll also be able to add support to Twitter-like APIs through external *.jar files, which will enable your app to make calls to the cloud.

3.1 Creating a user interface

A user interface in Android can be created in two ways, viz. using XML or writing Java code to develop the UI.

3.1.1 Declarative approach

Declarative user interfaces can be created using XML to declare what the UI will look like. It's similar to creating a web page using HTML. You use tags and specify which elements will appear on your screen. The advantage of this tool is that you can use "What You See Is What You Get" (WYSIWYG) tools, most of which are shipped with the Eclipse Android Development Tools (ADT) Extension.

However, XML only gives you the freedom to easily declare the look and feel of your user interface; it doesn't handle the user input well.

3.1.2 Programmatic approach

The programmatic approach for developing the Android user interface involves coding in Java. Android is more or less similar to Java AWT or Java Swing development. Hence, in order to create a button, you declare a button variable, create its instance, add it to a container and set the button properties such as color, text, size and background. You may have to write a code to declare the action of a button on click or hold; i.e. it involves considerable amount of coding in Java.

3.1.3 The right way

The best practice is to use the best of both the approaches. Use XML to declare all static elements of the user interface such as the layout, and the widgets etc. and switch to the programmatic approach to define the actions